

Incremental Learning in Network Traffic Management: A Fixed-Representation Rehearsal Approach

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The Problem

- **Traffic Drift** → Classifiers become outdated.
- **Full Retraining** → High computational and storage cost.
- **Naïve Fine-Tuning** → Catastrophic forgetting.

Current Solutions

- **Class-Incremental Learning** → Reduces forgetting but relies on costly replay/distillation mechanisms that conflict with resource-constrained deployments.
- **Fixed-representation combined with rehearsal** → Underexplored approach, restricted to packet-level scenarios.

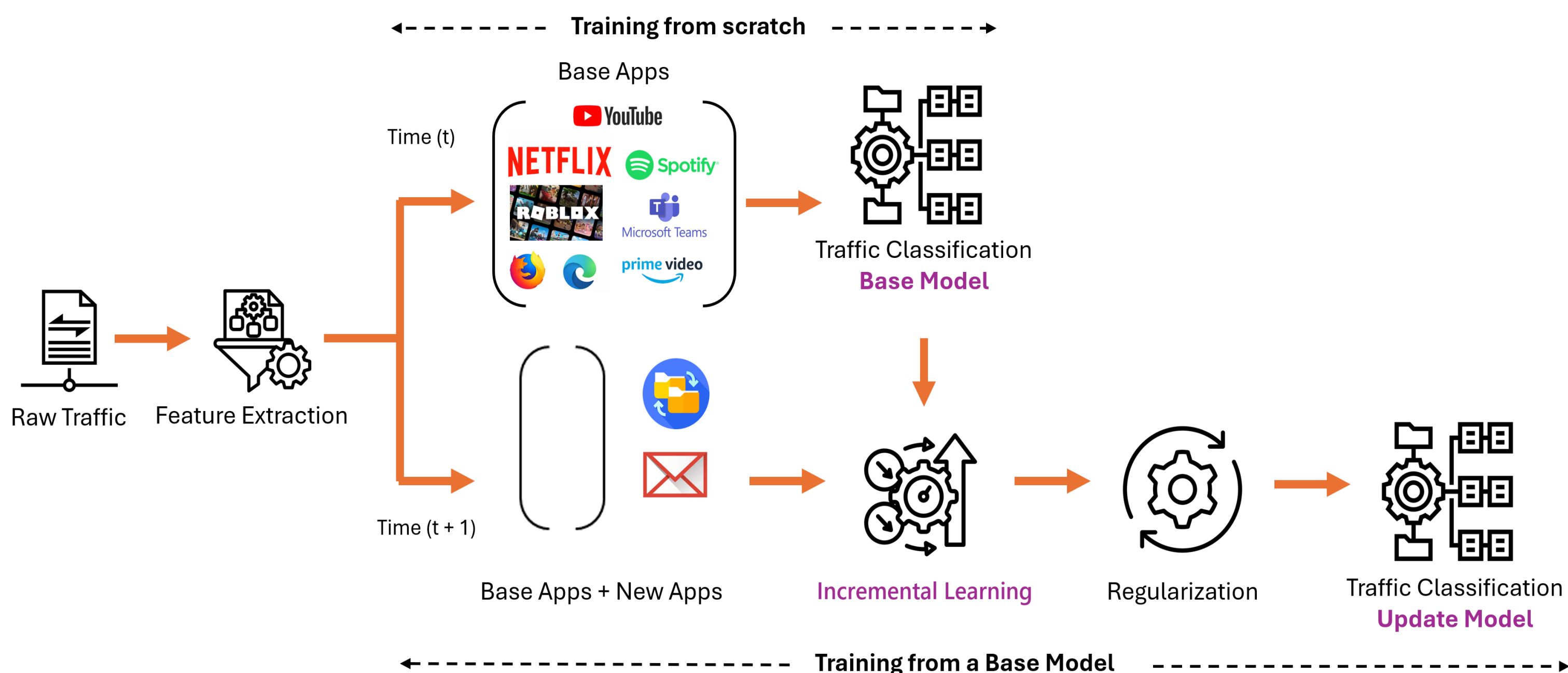


Fig. 1: Stages of the evaluated incremental-learning model — a frozen FCNN backbone with a 10% rehearsal buffer; only the classification head is retrained for the new class(es).

Our Contribution

- First dedicated evaluation (*to our knowledge*) of a **fixed-representation rehearsal** strategy at the **flow level**. Training cost & memory instead of maximum accuracy.
- **Freeze the whole feature-extraction backbone** (hidden layers 1–5); re-initialise and retrain **only the classification head**.
- Rehearsal buffer: **10% of the base training set** (stratified), replayed with all samples of the new class(es).
- Outcome: representational stability + fast updates that avoid catastrophic forgetting without complex pipelines.

Experimental Setup

Dataset: 46,729 labeled flows (VLC + VNAT) — 8 classes, 29 flow-level features (5-s windows), 80/20 train/test split.

Base model — 6 classes → Video Streaming, Roblox, Spotify, Teams, YouTube, Web.

Scenario A (6+1) → + File Transfer

Scenario B (6+2) → + File Transfer, + C2 (Command & Control).

FCNN: 29 → 2048 → 1024 → 768 → 512 → 256 → softmax (ReLU, BatchNorm, Dropout, L2; AdamW, lr 1e-4).

Training time

Wall-clock time to train the model
from scratch → incremental : reduction

7 classes → 6 + 1 classes : **1,696 s** → **133 s** : **92.16%**
8 classes → 6 + 2 classes : **1,872 s** → **192 s** : **89.72%**

Storage

Memory footprint of the training data
from scratch → incremental : reduction

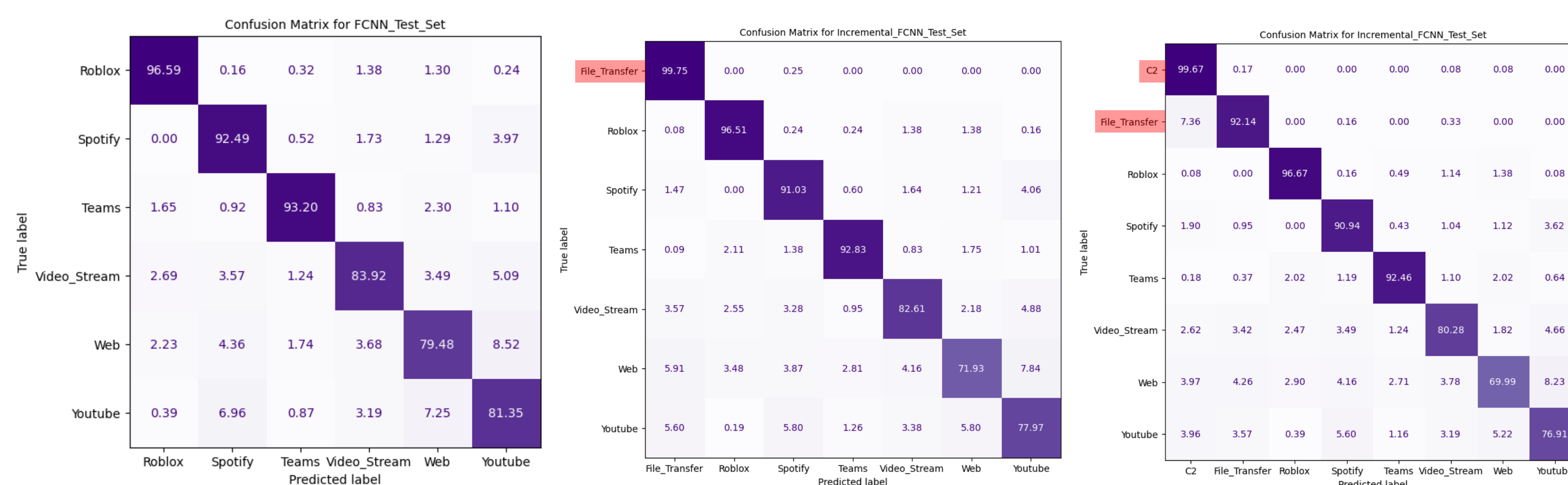
7 classes → 6 + 1 classes : **9.2 MB** → **2.1 MB** : **76.88%**
8 classes → 6 + 2 classes : **10.6 MB** → **3.4 MB** : **67.50%**

Forgetting

Accuracy lost on old classes when adding new ones
old-class accuracy, from scratch → incremental : gap

7 classes → 6 + 1 classes : **86.4%** → **85.8%** : **0.56%**
8 classes → 6 + 2 classes : **86.0%** → **84.9%** : **1.13%**

Results



(a) Base — 6 classes · 88.0%

(b) +1 class (6+1) · 87.9%

(c) +2 classes (6+2) · 87.7%

Fig. 2: Row-normalised confusion matrices (%): base 6-class model, then the incremental model after adding 1 and 2 new classes.

Next Steps

- **Long-Term Evaluation** → Assess performance across longer continuous-learning update sequences.
- **Scalable Scenarios** → Extend beyond 8 classes with imbalanced and evolving traffic arrivals.